

Hazard Analysis Physiotherapy Companion

Team 11, technically functional

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Table 1: Revision History

Date	Developer(s)	Change
September 29, 2025	Maham S, Matthew H	Initial table input
October 4, 2025	Matthew H	Added initial draft of HA
October 5, 2025	Maham S	Table additions
October 8, 2025	Matthew H	Table mods
October 9, 2025	Maham S	Final edits

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1 Introduction

The team’s definition of a hazard is an “unsafe” condition that may lead to an unintended event that causes an undesirable outcome [Ippolito and Wallace, 1995]. In relation to software development, hazards can be understood as the underlying design decisions that result in potential harm. If these hazards remain unaddressed prior to development, they may become a source of risk. Hazards can include technical failures, privacy/security, usability and accessibility risks.

The project’s goal is to develop a tool that provides guidance for a physiotherapist’s prescribed home exercises. The user’s form is analyzed and prompts for adjustments are given. While this can increase the rate and efficacy of recovery, it introduces several hazards. [See the Section 3 in the Problem Statement document for more on project goals.](#) The purpose of this document is to identify those hazards, outline assumptions and propose solutions to limit their presence and their frequency.

2 Scope and Purpose of Hazard Analysis

This document outlines the hazards that may occur while operating the tool. The key areas for scope include:

- **Safety-Critical Technical Hazards:** This involves failures in the underlying logic or analysis that directly impacts patient’s safety. This includes incorrect identification of exercise planes, improper landmark detection, or inaccurate joint angle calculations. These failures may result in misleading real-time feedback, potentially causing the user to perform unsafe movements which may lead to re-injury.
- **Operational Technical Hazards (Application Performance):** Hazards that affect the application’s functionality without risk of patient injury. These include system latency, inability to process footage due to hardware constraints, or database synchronization errors. While these impact the user experience and data integrity, they do not cause immediate harm.
- **Usability Hazards:** Usability ensures that the tool has the potential for successfully being adopted. All of the inputs to the system, such as the recording of the exercise, are generated from the user. A user-friendly interface guarantees ease of use, which encourages utilization of this tool. On the other hand, incorrect, on-screen prompts may cause confusion while the user performs the exercise. This may reduce usability and lead to ambiguous guidelines.
- **Privacy/Security Hazards:** Unnecessary data collection and storage increases the potential of unauthorized access and misuse of personal information. Such situations, like data breaches, may lead to catastrophic outcomes, which will result in lower adoption rate.

The purpose of this document is to identify risks and their associated impact to determine ways to mitigate them and identify any future occurrences

3 System Boundaries and Components

To better understand the application, we divided the application into 7 different modules. These modules are not necessarily separate entities but they are different components of the application that work together.

- Module 1: Authentication and Account Management
 - Manages patient and physiotherapist accounts, including registration, authentication through Firebase Authentication. It ensures that only authorized patients can access personal health data and exercises assigned by their physiotherapist.
- Module 2: Organization Linking and Access Codes
 - Facilitates connection between a patient and their physiotherapist. This module also handles the validation of unique access codes for the physiotherapist to provide to the patient.
- Module 3: Exercise Program Management
 - Manages a library of exercises assigned by the physiotherapist to the patient. This module handles how and what the patient receives information about the exercises.
- Module 4: Video Capture and Review
 - This module is responsible for the real-time video capture of the patient from the device's camera. The video is then shown on the device for live feedback and repetition counting.
- Module 5: Computer Vision and Form Analysis
 - Uses computer vision library to analyze the patient's form during exercise. These metrics are then evaluated for landmarks at different joints then evaluates the movement against the ones done prior to ensure that each rep is performed correctly.
- Module 6: Reporting and Dashboards
 - Consolidates the weekly data of the patient into digestible reports. This module also surveys the patient for subjective data such as Rate of Perceived Exertion (RPE) and pain levels. This module also provides a dashboard for the physiotherapist to view the patient's progress and make adjustments to the exercise program as needed.
- Module 7: Data Security and Privacy

- Ensures the safety and security of patient data with low coupling between modules.

These system boundaries are included within the scope of the hazard analysis since they make up the core functionality of the application.

4 Critical Assumptions

- Assume that the user only has mobility issues pertaining to the knee area and with single joint activity.
 - Moreover, it is assumed that the user will perform an exercise that resembles the exercise.
- Assume that only one user is within the frame of the mobile device.
- Assume that the user is receptive to feedback.
- Assume that the user is honest when providing their subjective data such as RPE and pain levels.
- Assume that the user has a stable connection to the internet.
- Assume that the user is undergoing physiotherapy and was advised to perform the exercise at home.
- Assume that the device being used meets the minimum requirements for our system to function.
- Assume that the physiotherapists using the application are licensed.

5 Failure Mode and Effect Analysis

Table 2: Failure Modes and Effects Analysis

Design Function	Failure Modes	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref.
Authenticate user	Unautho- rized access to different patient data	Breach in privacy to personal in- formation	A. Having a weak pass- word	A. User are re- quired to have to create certain password require- ments to create an account and sign in.	A. FR1.4 B. SR1 C. NFR 2.3	H1-1
			B. No form of 2FA to pro- tect personal data	B. Users are re- quired to enter an email when creating account for 2FA		
			C. Applica- tion keeps user signed in constantly	C. The system will log the user out after 10 minute of in- activity		

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Design Function	Failure Modes	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref.
Authenticate user	Unable to access user profile	User unable to log into app	A. Technical failure (eg database access down) B. Too many failed login attempts	A. Implement pre-release verification of external library and database connectivity B. Implement a change password option by sending email	A. SR2 B. SR3	H1-2
Detecting user's body (input video stream)	Unable to detect user's starting form (stationary)	App is unable to provide feedback	A. The environment around the user interferes with the system's detection B. App is unable to use the camera	A. Prompt the user to change their environment to a controlled environment without obstructions to the camera B. Close other apps that may be using camera	A. SR4 B/D. SR5 C. SR6 E. FR2.5	H2-1

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Design Function	Failure Modes	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref.
			C. Device with insufficient charge	C. Prompt the user that they need sufficient charge to proceed with the analysis		
			D. App does not have camera access permission	D. Grant permission for app to access camera		
			E. Application does not have sufficient data to analyze user	E. Application support retry attempts for the user's form if not aligned or has obstructions		
Detecting user's body (input video stream)	Unable to detect user's starting form (stationary)	Application is unable to provide accurate feedback	A. The environment around the user interferes with the system's detection	A. Refer to H2-1A	A. SR4 B. SR2	H2-2

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Design Function	Failure Modes	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref.
			B. The system places markers incorrectly	B. Refer to H1-2A		
Detecting user's body (input video stream)	Unable to detect user's starting form (stationary)	Limited, incorrect or no footage provided for processing different body parts	A. The environment around the user interferes with the system's detection B. Device with insufficient charge C. Application does not have sufficient data to analyze user	A. Refer to H2-1A B. Refer to H2-1C C. Refer to H2-1E	A. SR4 B. SR6 C. FR2.5	H2-3

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Design Function	Failure Modes	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref.
Analyzing exercise form (monitoring)	Unable to calculate metrics properly	May mislead the user	A. Incorrect placement of the body during exercise B. Environment is cluttered and there are interferences during analysis C. Application does not have sufficient data to analyze user	A. Provide some sort of outline on the screen where the user has to position themselves to make each view standardized B. Refer to H2-1C C. System supports retry attempts if insufficient data	A. SR7 B. SR4 C. SR8	H3-1
Analyzing exercise form (monitoring)	Unable to calculate metrics properly	Causes feedback to be incorrect and may satisfy the wrong requirements	A. Incorrect placement of the body during exercise	A. Refer to H3-1A	A. SR7 B. SR4	H3-2

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Design Function	Failure Modes	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref.
			B. Environment is cluttered and there are interferences during analysis	B. Refer to H2-1A		
Analyzing exercise form (monitoring)	Unable to calculate metrics properly	App is unable to provide feedback	A. Incorrect placement of the body during exercise B. Environment is cluttered and there are interferences during analysis C. Application does not have sufficient data to analyze user	Refer to H3-1	A. SR7 B. SR4 C. SR8	H3-3

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Design Function	Failure Modes	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref.
Real-time feedback	Feedback is slow and does not provide it to the user well	Users cannot correct form and delay in verification of the new form may cause poor performance of the movement	A. Implementation of metrics maybe delaying how prompts are generated B. Application does not have sufficient data to analyze user	Review how the metrics are being generated and when prompts are being given to the user (i.e. running average but it takes many data points) Refer to H3-3C	A. NFR1.3 B. SR8	H4-1
Analyzing exercise form (monitoring)	Unable to calculate metrics properly	System provides the users with incorrect feedback outside of their physiotherapist's recommendations	System does not know the limitations set by the physiotherapist	Allow a manual input to set daily/weekly limit on increases in-load	A/B. SR9	H4-2

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Design Function	Failure Modes	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref.
				B. Allow a manual input to set daily/weekly limit on increases on mobility		
Analyzing exercise form (monitoring)	Confusing prompts (jargon)	User may not be able to perform the exercise as intended	Prompts are confusing and use jargon in their instructions	A. The system provides a tutorial of the movement during feedback B. Refer to H4-2 C. Design for accessibility (audio cues and visual cues)	A. SR10 B. SR9 C. NFR3.1	H4-3
Analyzing exercise form (monitoring)	Confusing prompts (jargon)	User may harm themselves as a result of the technical jargon	Prompts are confusing and use jargon in their instructions	Refer to H4-3	A. SR10 B. SR9 C. NFR3.1	H4-4

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Design Function	Failure Modes	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref.
Analyzing exercise form (monitoring)	Confusing prompts (jargon)	User might not know what to correct and become frustrated thus not use the app	Prompts are confusing and use jargon in their instructions	Refer to H4-3	A. SR10 B. SR9 C. NFR3.1	H4-5
Data Management (database)	The system pulls incorrect patient data	This can lead to a security breach and improper access of other user data	Prompts are confusing and use jargon in their instructions	A. The system provides a tutorial of the movement during feedback	A. NFR2.1	H5-1
Data Management (database)	The system pulls incorrect patient data	This can affect how feedback is given and stored for future	System incorrectly takes from the wrong userID	Refer to H1-2A	A. SR4	H5-2
Data Management (database)	The system pulls incorrect patient data	There are duplicate entries	Refer to H5-1A	Refer to H5-1A	A. NFR2.1	H5-3

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Design Function	Failure Modes	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref.
Data Man- agement (database)	Unable to access user profile	System is unable to log in user	Refer to H1-2	Refer to H1-2	A. SR2 B. SR3	H5-4

6 Safety and Security Requirements

SR1: The system shall enforce a 2FA when creating an account and logging in. The system will ensure 2FA through the email address of the user stored on the database.

Rationale: Users who do are not properly authenticated could result in privacy breaches to personal and healthcare information.

Risk: Medium

Reference: H1-1

SR2: The system shall test the connectivity to external libraries and database connectivity prior to release.

Rationale: This can result in functionality issues with the application and also obtaining data from the incorrect account.

Risk: High

Reference: H1-2, H2-2, H5-2

SR3: The system shall include an option to reset the password if too many (5) attempts are made with incorrect login.

Rationale: Inability to access the user's profile can result in not having access to the application's functionality.

Risk: Medium

Reference: H2-1, H5-2

SR4: The system shall prompt the user at the start of video streaming if there are obstructions or low-light levels and give the user a chance to change their environment prior to analysis.

Rationale: System will be unable to decipher form if the surrounding environment is not correct.

Risk: High

Reference: H2-1, H2-2, H2-3, H3-1, H3-2, H3-3, H5-2

SR5: The system shall prompt the user if the application does not have access to the camera or if there is an alternative program that is using the camera.

Rationale: This can result in an inability to detect the user's form and the application will have nothing to process.

Risk: Medium

Reference: H2-1, H3-2

SR6: The system shall monitor the mobile device's battery life and prompt the user if it falls below a certain threshold to charge the device.

Rationale: The system cannot function without proper access to power.

Risk: Low

Reference: H2-1, H2-3

SR7: The system shall provide a guideline overlay when setting up the mobile

device so they are in-frame of the view and correct orientation.

Rationale: In order to align the user to the camera and detect the user's posture throughout the exercise.

Risk: High

Reference: H3-1, H3-3

SR8: The system shall support retry attempts if the patient's form has insufficient data.

Rationale: The purpose is to provide analysis and if insufficient data is collected then it will notify the user to retry.

Risk: High

Reference: H3-1, H3-3, H4-1

SR9: The system shall allow for physiotherapists to input daily/weekly limitations on the metrics measured by the application.

Rationale: Exceeding the limits may result in improper or unsafe form and cause injury.

Risk: High

Reference: H4-2, H4-3, H4-4, H4-5

SR10: The system shall provide an option to view the motion on an internal page on the application to help guide the motion during analysis.

Rationale: Without this requirement, it may be difficult to replicate suggestions in words, having a visual cue may improve accessibility.

Risk: Low

Reference: H4-3, H4-4, H4-5

7 Hazard Analysis Traceability Matrix

The following matrix showcases the dependencies between the hazards listed and all of the requirements. This can be seen in our traceability matrix linked in the URL below, which will take you to the excel spreadsheet with the matrix:

[Hazard Analysis Traceability Matrix](#)

8 Roadmap

This section outlines which safety requirements that will be implemented within the span of the capstone timeline. It was decided that the measure of severity as well as likeliness of the hazard occurring will dictate which safety requirements are implemented. Above all, the scope of the project is also considered and the outlook on developing different requirements. As a team we have acknowledged that if everything could be implemented during the timeline of the capstone than it would result in a better final product but the limitation on time and

developers were taken into account.

The safety requirements that will be part of the capstone time-line include:

- SR2 - This is critical to the core functionality of the application and will not be functional without this requirement.
- SR3 - This will be include as it is a relatively simple requirement to implement and has a risk level of medium.
- SR4 - This is within scope as if the user is not within a suitable environment, the application may not be able to provide feedback or provide inaccurate feedback.
- SR5 - This will be within scope as the application will not perform core functionality if the camera is not accessible.
- SR7 - This will be included to promote ease of user interaction with the application.
- SR8 - This will be included as there will be no output from the application if there is insufficient data to analyze.

The requirements that will be implemented in the future include:

- SR1 - This will be outside the capstone timeline since currently it is not integral to the minimum viable product.
- SR6 - This will be outside the capstone timeline as it is a more advanced feature that is not critical for the minimum viable product.
- SR9 - This will be outside the capstone timeline as there must be additional value add from a business standpoint prior to implementing this safety requirement. Additionally, after discussing with our supervisor, this safety requirement does not apply to our initial case of post-operative knee exercises as mobility is necessary during the early stages of recovery.
- SR10 - This will be outside of the capstone timeline as this does not pertain to the core functionality of the program for the capstone and can be seen as an additional safety requirement.

Appendix — Reflection

[Not required for CAS 741 —SS]

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. Which of your listed risks had your team thought of before this deliverable, and which did you think of while doing this deliverable? For the latter ones (ones you thought of while doing the Hazard Analysis), how did they come about?
4. Other than the risk of physical harm (some projects may not have any appreciable risks of this form), list at least 2 other types of risk in software products. Why are they important to consider?

Matthew - Reflection

1. I think what well in this deliverable had to have been our communication. This deliverable had a lot more separate sections that required each of us to be subject matter experts on, for example I primarily worked on the HA while others primarily did the SRS. Being able to communicate effectively allowed us to divide work while also being aware of how the other deliverable was doing and if any help was needed there.
2. I think a pain point that we experienced this deliverable was the idea of the application we were building. While we wanted to keep things general, the group found it hard to move forward without a clear idea of the project and a more granular detail. This resulted in each member having their own interpretation of the project each with slight differences that it affected consistency within documents and sections. The way that we resolved this was by identifying that there was part of the application that we had discrepancy on. From there we scheduled a meeting to determine the general flow of the program. This allowed each of us to voice opinions

while also understanding each team members point of view of the project. I would confidently say that since then we resolved those issues.

3. The risks that the team had thought of before this deliverable were very basic, it wasn't until this deliverable where we were able to add requirements to the system and then identify hazards. The ones that we thought of from the SRS are identified in the FMEA table labelled FRX.X or NFRX.X. The ones that we came about during the hazard analysis were the ones labelled SRX.X. This came about by putting ourselves in the shoes of the stakeholders and trying to understand issues that they might have. Furthermore, we had asked our advisor (Kevin), who is a physiotherapist, his perspective and how he would also view pain points. This allowed us to develop a list of hazards and thus requirements to address for the application.
4. Two other risks are technical and security risk in software products. These are important to consider as if they go unchecked then they can lead to catastrophic consequences. Technical risk for example may lead to unintended functionality and serve as a disconnect between what wanted to get implemented and what was implemented. Security risk, especially when collecting data and storing it, can result to data breaches and lead to the failure of protecting sensitive data. These both can result in a loss of trust when using different software and poor rates of adoption. Furthermore they can contribute massively towards cost of the project if a hazard would occur. These considerations must be taken into account for the health of the software and success.

Maham - Reflection

1. The process of ensuring that all team members are on the same wavelength in terms of what the application is supposed to do was very interesting. Throughout the deliverable, everyone communicated their understanding well and I was impressed with how quickly we were able to sync up. In addition, all members were able to hop on and assist anyone that needed any. This was only possible because we would conduct team meetings very often and check in on where we were with our assignment, That helped stay on track with the schedule as well.
2. I think the most important pain point was the constant juggling between priorities for each member. Everyone had assignments and midterms to complete, and that required very nuanced scheduling (to do list). I work by dividing up everything I have to do into small chunks and switch between tasks after a certain amount of time. I think most members on my team were also doing something similar to this and did not leave the bulk of their work for last minute.
3. I think the risks we thought of before working on the deliverable were quite vague and unspecific (eg. user data security). Hazards and risks

were thought of, but not solution or fixes. Once we started completing the deliverable, there were some processing hazards that we had not thought of come to light. Thinking of all the things that are possible to go wrong made it imperative to address, and we started developing fixes, whether they were similar to any others already present in the table etc.

4. As mentioned above, security risks should be considered. Patients are providing the application important and private information (e.g. videos) and they should be handled ethically. A second risk that comes to mind is the potential of the application being misused. In the context of our application, we designed it so any of the physiotherapists that are registering on the app are licensed. If not, then anyone posing as a physiotherapist can potentially use the application and create liability issues for the developers.

Vaisnavi - Reflection

1. Communication throughout the team is something that was very present throughout this deliverable. Matthew was the first one to start the HA document and kept us in the loop throughout. This helped when I had to hop on for the traceability matrix as it aided in providing insight on the concept of hazard analysis and made it seem less daunting.
2. It was hard at first to visualize how the traceability matrix would be mapped out and what it would consist of. However, after talking to the professor, the TA and my teammates, it allowed myself to envision clearly what the traceability matrix required and what I would be comparing to one another.
3. The risks the team brainstormed beforehand were very broad. We thought of hazards such as issues related to the security of our data but nothing was solidified. Only once we started to work on the document was when the remaining latter ones started to come about. As we started to have more of a grasp on our project as a whole, this was when the remaining ones began to come to light.
4. In terms of types of risks that come about in software products, one principal one previously mentioned which is a very common one is data security risk. Any access to data that is not authorized is very harmful and can cause numerous issues and leaks of personal information. Another key risk is any sort of failures that happen in regards to the reliability of the system. This could involve something like a system crash, which would ultimately impact the user's activity and their overall user safety as well.

Cieran - Reflection

1. The hazard analysis was undertaken mainly by my colleagues. However, there was a lot of communication between the two teams in terms of

keeping the files in line with one another. Specifically, communication went well regarding the assumptions which were critical for the system definitions and the safety requirement pertaining to the risks and hazards included in the hazard analysis.

2. There were misalignments in understanding of the project and in certain senses what requirements met. After team meetings and clear discussions, the ambiguity has been hammered out of both the team and the documents to put the best foot forward on the system from the get-go.
3. Risks thought of before the project for the system particularly included, but were not limited to, data privacy leaks and healthcare advice misinterpretation. Data privacy is a huge thing, users want to know what their data is being used for, where it goes, who sees it, and why all of this has to be done. Keeping data private and out of sight from those who don't need to see it is a big thing to consider within the system. The risk of that data escaping is something that has been considered the entire time. As for healthcare advice, the system isn't replacing healthcare professionals and should not try to be one nor mimic the actual healthcare that professionals give, the system is intentionally only meant to provide recommendations. Staying away from healthcare involvement is a risk for the system to avoid liabilities.
4. Considerable risks within software products include security vulnerabilities and insufficient testing/poor code quality. Security vulnerabilities, as previously discussed for this system, can expose private user data, but vulnerabilities in other platforms can be far more detrimental. Developers have to be very careful with their security measures when interacting with sensitive user data, especially when data which is leaked could drastically affect the user outside the contexts of the system. Insufficient testing and/or poor code quality can make a poor implementation into a risk for user devices and performance. If the code is poorly optimized or even broken in some places, there's a risk of causing issues with a user's device.

Eman - Reflection

1. Something that went well while writing this deliverable is the coordination between members who were working on the Hazard analysis document and the SRS requirements. As there is a strong connection between the two, good communication was required to ensure consistency throughout.
2. One pain point I experienced in this deliverable was understanding to what extent the details would be beneficial, and at what point would it be detrimental to narrow down the requirements further. I was afraid of overcommitting responsibilities without knowing our capacity and capabilities for the future. This was resolved by outlining requirements that may be stretch goals. This way, instead of over committing without enough knowledge we are still noting down potential scope for our project.

3. We had previously identified technical risks like computer vision accuracy and data security concerns, as these were obvious given the project’s nature. However, while writing requirements, we discovered safety-critical risks such as patients injuring themselves from incorrect form when exercising unsupervised at home, which emerged from discussions about the fit criteria for real-time feedback and led to PR-SCR 1. We also identified usability risks for elderly or non-technical users during the accessibility requirements section, realizing that poor usability could lead to treatment non-adherence and worsen health outcomes.
4. Privacy and data breach risks are critical in healthcare applications because unauthorized access to patient health information can lead to identity theft, discrimination, legal liability, and loss of patient trust in the healthcare system. Treatment efficacy and clinical outcome risks are important because inaccurate computer vision analysis or poor feedback could lead to ineffective treatment, prolonged recovery times, or patients developing compensatory movement patterns that cause secondary injuries, ultimately undermining the therapeutic goals of the application.

References

Laura M. Ippolito and Dolores R. Wallace. *A Study on Hazard Analysis in High Integrity Software Standards and Guidelines*. 1995. URL <https://nvlpubs.nist.gov/nistpubs/Legacy/IR/nistir5589.pdf>. U.S. Department of Commerce, Technology Administration, National Institute of Standards and Technology.